

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. CSE(AI&ML) / AI&DS

Semester : VII

Subject Code & Name: BTAIC701 Natural Language Processing

Max Marks: 60

Date:02-01-2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

- Q. 1 Solve Any Two of the following. 12**
- A) What is Natural Language Processing? Discuss with some applications. Understand 6
- B) What is the purpose of text normalization in natural language processing and Explain the three common tasks in text normalization? Understand 6
- C) Explain the "Minimum Edit Distance Algorithm" and compute the minimum edit distance between EXECUTION and INTENTION Apply 6
- Q.2 Solve Any Two of the following. 12**
- A) What is a language model? Write note on N-Gram language model. Understand 6
- B) Find the probability of test sentence given below if a bigram model is used on the following training sentences Apply 6
- Training data:**
- 1.The Arabian Knights
- 2.These are the fairy tales of the eat
- 3.The Stories of the Arabian Knights are translated primary languages.
- Test Data: The Arabian Knights are the fairy tales of the eat**
- C) Explain vector space model of information retrieval 6
- Q. 3 Solve Any Two of the following. 12**
- A) Perform parsing using simple top down parsing for the sentence "The dogs cried" using the grammar given below: Apply 6
- S->NP VP
- NP->ART N
- NP->ART ADJ N
- VP->V

VP->V NP

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|--|-------------------|----------|
| B) Show working of NER with appropriate example. | Understand | 6 |
| C) Illustrate maximum entropy principle formula with example. | Understand | 6 |

Q.4 Solve Any Two of the following. **12**

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|---|-------------------|----------|
| A) Write a short note on importance of sentiment analysis. | Understand | 6 |
| B) Apply the Logistic Regression model to perform text classification. | Apply | 6 |
| C) What is Text Classification? Explain How Does Text Classification Work? | Understand | 6 |

Q. 5 Solve Any Two of the following. **12**

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|---|-------------------|----------|
| A) Explain text summarization and multiple document text summarization with neat diagram | Understand | 6 |
| B) Explain the simple RNN with its application | Understand | 6 |
| C) What is textual entailment in NLP? Apply it on question answering application and simplify its working with suitable example. | Apply | 6 |

***** End *****